

Interference-aware multi-dimensional resource scheduling for beam hopping in NGSO satellite constellations

Shengjun Guo^{a,b}, Kai Han^{a,b}, Lu Li^b, Wenbin Gong^{b,*}

^a University of Chinese Academy of Sciences, Beijing 100049, China

^b Innovation Academy for Microsatellites of Chinese Academy of Sciences (IAMCAS), Shanghai 201304, China

Received 3 August 2025; received in revised form 6 December 2025; accepted 20 December 2025

Available online 26 December 2025

Abstract

The rapid advancement of Non-Geostationary Orbit (NGSO) satellite constellations is propelling the evolution of next-generation satellite internet toward seamless global coverage and high-speed connectivity. Beam hopping (BH), with its ability to flexibly steer spot beams in a time-division manner, has emerged as a key technique for matching spatiotemporally dynamic traffic demands with limited on-board resources. However, dense beam deployment induces severe co-channel interference (CCI), and the coupling of multi-dimensional resources further complicates the scheduling process. To address these challenges, this paper introduces an integrated, efficient, and readily deployable interference-aware cooperative resource scheduling framework for multi-satellite BH systems. The original optimization problem is decomposed into three subproblems: satellite–cell association, joint illumination pattern and bandwidth allocation, and power optimization. Satellite-cell association is determined by simulated annealing (SA) algorithm to achieve inter-satellite load balancing. Furthermore, the beam illumination pattern is designed via integer linear programming (ILP), while the beam-frequency allocation is jointly optimized based on interference graph and graph coloring theory. Finally, a bandwidth-aware water-filling algorithm is developed to determine the power allocation for each active beam. Simulation results demonstrate that the proposed approach achieves a 94.2% improvement in inter-satellite load balancing. Under high traffic intensity, it delivers significantly higher system throughput and reduces average delay by over 16% compared to baseline schemes. Moreover, the approach consistently maintains high user satisfaction and ensures service fairness across all cells, with effective tail-performance control under varying traffic intensities.

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Keywords: Beam hopping; Resource allocation; Load balancing; Interference avoidance; NGSO constellations

1. Introduction

The accelerated evolution of Non-Geostationary Orbit (NGSO) satellite internet systems has ushered in a new era of global connectivity, serving as a critical complement to terrestrial networks while addressing the need for ubiquitous broadband (Chen et al., 2020). Fueled by sharply

lower launch costs and large private investments, emerging mega-constellations such as SpaceX's Starlink, Eutelsat OneWeb, Amazon's Kuiper and Telesat Lightspeed are deploying large numbers of satellites in Low Earth Orbit (LEO) to provide high-speed internet access, especially in remote and underserved regions (Al-Hraishawi et al., 2022; del Portillo et al., 2019). To facilitate integration of satellite systems into future wireless networks, standardization bodies including the 3rd generation partnership project (3GPP) have incorporated Non-Terrestrial Networks (NTN) into their frameworks, positioning NGSO constellations as a key enabler for beyond 5G (B5G) and future

* Corresponding author.

E-mail addresses: guoshengjun20@mails.ucas.ac.cn (S. Guo), han-kai20@mails.ucas.ac.cn (K. Han), lilu16@mail.ustc.edu.cn (L. Li), spg3@163.com (W. Gong).

6G communications (Chen et al., 2020; Lin et al., 2021; Sheng et al., 2022). Nevertheless, the rapid movement and constrained resources of LEO satellites, together with the inherently heterogeneous distribution of ground traffic over space and time, pose major obstacles to achieving dynamic and optimal matching between uneven demand and limited resources. While multi-beam satellite (MBS) systems are designed to address such challenges by enhancing capacity through the generation of multiple high-gain narrow beams, their static and uniform beam service patterns often lead to inefficient utilization of satellite resources (Li et al., 2018).

Beam hopping (BH) technology leverages the agile beamforming capability of phased array antennas to dynamically adjust beam direction and duration time, offering a flexible solution to accommodate spatiotemporally varying traffic demands. The inherent flexibility of BH in allocating multi-dimensional resources, encompassing illumination pattern design (time and spatial directions), bandwidth, and power allocation, has motivated significant research into advanced resource allocation algorithms for BH systems. Liu et al. (2018) proposed an iterative algorithm to maximize system capacity, demonstrating strong performance under non-uniform traffic distributions. To jointly optimize beam pattern and power allocation, a low-complexity algorithm based on potential game and interior-point method was proposed (Shuang et al., 2024), achieving promising performance in both throughput and fairness. In addition, given the NP-hardness of the BH resource optimization problem, heuristic algorithms are commonly employed to obtain near-optimal solutions with reasonable computational complexity (Alegre et al., 2011; Anzalchi et al., 2010). Guo et al. (2022) formulated a multi-objective optimization framework to jointly maximize system throughput and minimize packet loss and delay, where an improved genetic algorithm (GA) was proposed to jointly optimize the allocation of beam bandwidth and power. To accelerate convergence, Wei et al. (2022) proposed an improved artificial bee colony (ABC) algorithm that optimizes the BH pattern with the dual objectives of capacity enhancement and service fairness. A dynamic beam illumination strategy integrating a low-complexity heuristic algorithm and selective precoding was introduced by Chen et al. (2023) to minimize inter-beam interference, achieving superior performance in terms of user satisfaction and system throughput.

However, due to the limited adaptability of heuristic methods in highly dynamic environments, deep reinforcement learning (DRL) has emerged as a promising alternative for adaptive and scalable BH optimization. Du et al. (2022) and Ran et al. (2024) optimized beam hopping and power allocation to simultaneously improve system throughput and resource utilization efficiency. Xu et al. (2023) proposed a DRL-based algorithm to jointly optimize the BH scheduling and coverage control which flexibly uses three degrees of freedom of time, space and beam coverage radius. Lin et al. (2022) proposed a multi-

agent DRL algorithm to jointly optimize dynamic beam pattern and bandwidth. Moreover, a novel multi-action selection scheme based on double-loop learning was proposed by Hu et al. (2020) to address the action space dimensionality problem, enabling simultaneous multi-objective optimization and adaptive resource allocation according to user demands and channel conditions.

Nevertheless, the above studies are primarily limited to single-satellite scenarios. As NGSO constellations continue to scale up, resource scheduling becomes increasingly complex, particularly due to the exacerbation of co-channel interference (CCI). Consequently, growing attention has been directed toward multi-satellite BH scheduling. Lin et al. (2023, 2024, 2025) proposed multi-satellite BH resource scheduling schemes for NGSO systems, which jointly consider load balancing and interference avoidance. Leveraging linear programming techniques, Pachler et al. (2024) present flexible and efficient algorithms for beam frequency assignment in mega-constellation-based multi-beam satellite systems, effectively mitigating interference while enhancing system capacity and improving power efficiency. Besides, Zhao et al. (2024) investigated a heterogeneous system where LEO satellites at different orbital altitudes provide flexible beam coverage, and proposed a user satisfaction-aware cell-satellite association strategy to achieve inter-satellite load balancing. A multi-agent DRL-based cooperative BH framework was proposed by Wang et al. (2025), jointly optimizing scheduling, resource allocation, and ISL coordination to enhance throughput and delay under large-scale LEO constellations. Wang et al. (2024) introduced a BH and frequency allocation framework based on DRL and majorization-minimization to improve throughput and delay fairness in multi-satellite LEO systems. Moreover, Wen et al. (2024) proposed a multi-agent DQN algorithm where agents collaborate by sharing global states and rewards, effectively enhancing system throughput under interference isolation. While contributing to algorithmic research, Samuel Martínez Zamacola and Rodríguez-Orsorio (2025) extends its scope to explore the architectural constraints and economic trade-offs in practical beam-hopping deployments.

Despite these advancements, most existing studies reduce problem complexity by employing full-frequency reuse and mitigating interference primarily through spatial separation (Lin et al., 2023; Wen et al., 2024; Hu et al., 2020). However, this strategy restricts the concurrent service of adjacent high-demand cells, limiting its applicability in scenarios requiring dense and flexible coverage, such as emergency communications. Some DRL-based approaches attempt to address interference by incorporating it as a weighted term in the reward function (Lin et al., 2024), which may not provide reliable interference suppression due to the absence of explicit coordination mechanisms. Moreover, current research predominantly focuses on optimizing the BH pattern, while joint optimization of multi-dimensional resources remains underexplored. In

particular, bandwidth and power allocations are often strongly coupled, yet many works only consider power allocation in isolation (Ran et al., 2024; Shuang et al., 2024). Additionally, existing optimization efforts have primarily prioritized throughput and delay, with relatively little attention devoted to ensuring service fairness across ground cells.

Motivated by the aforementioned concerns, this paper proposes a flexible interference-aware multi-satellite BH multi-dimensional resource scheduling algorithm. Our primary goal is to develop a lightweight, stable, and practically deployable scheduling framework for joint multi-dimensional resource optimization. In order to make the original problem more tractable, this paper decomposes it into three subproblems, namely satellite-cell association problem, interference-aware joint beam illumination and bandwidth optimization problem and power allocation problem. The proposed approach can effectively balance inter-satellite loads, improve system throughput, and mitigate both intra-satellite and inter-satellite interference. The main contributions and innovations of this work are summarized as follows:

- Formulation of the multi-satellite BH scheduling problem, aiming at maximizing the system throughput and balancing loads across satellites. For the purpose of effectively solving the NP-hard problem, it is decoupled into three subproblems.
- Presentation of a suite of low-complexity algorithms to address the decomposed subproblems. Firstly, a satellite-cell association strategy based on simulated annealing (SA) algorithm is developed to achieve inter-satellite load balancing. Secondly, an integer linear programming (ILP)-based illumination pattern is optimized for throughput maximization, followed by a graph-based, interference-aware bandwidth allocation algorithm to eliminate interference. Thirdly, a bandwidth-adaptive power allocation problem is formulated and solved using a water-filling strategy to enhance resource utilization under power constraints.
- Extensive simulations are conducted under varying traffic intensities. Realistic uneven traffic distributions are modeled based on population density and typhoon trajectory data. Numerical experiments validate the superiority of the overall scheduling framework compared with different benchmarks in terms of load balancing, system throughput, average delay, satisfaction rate and user fairness.

The remainder of this paper is organized as follows. Section 2 introduces the system model and formulates the multi-satellite beam hopping optimization problem. Section 3 decomposes the original problem into three subproblems and presents corresponding solution methods. Section 4 provides comprehensive simulation results and performance evaluations. Finally, Section 5 concludes the paper and summarizes the main contributions.

2. System model and problem formulation

2.1. System model

In this paper, we consider the forward link transmission of a multi-beam LEO constellation as shown in Fig. 1, with N_s satellites providing services for N_c ground cells. The satellite set is denoted as $\mathcal{N}_s = \{i|i = 1, 2, \dots, N_s\}$ and the cell set is $\mathcal{N}_c = \{n|n = 1, 2, \dots, N_c\}$. Assume that each single satellite can cover N_m cells, denoted as $\mathcal{N}_m = \{m|m = 1, 2, \dots, N_m\}$. It is worth noting that some cells may be covered by multiple satellites at the same time, which is one of the typical characteristics of NGSO systems. Each satellite can simultaneously generate up to J beams, each of which can be flexibly configured with bandwidth and power, and provide services to the cells it covers in a time-division manner by BH technology. Define the spotbeam set as $\mathcal{J} = \{j|j = 1, 2, \dots, J\}$. The minimum dwell time of a beam is called a time slot, and a BH period—formally defined as the set $\mathcal{T} = \{t|t = 1, 2, \dots, T\}$ —comprises T consecutive such slots.

The cells illuminated simultaneously by a satellite during a specific time slot are referred to as its illumination pattern. Define the illumination pattern of the entire constellation as $\chi_t = [\chi_t^i|i = 1, 2, \dots, N_s]$, in which χ_t^i represents the illumination pattern of satellite i in time slot t . Specifically, $\chi_t^i = [\chi_t^{i,1}, \chi_t^{i,2}, \dots, \chi_t^{i,n}, \dots, \chi_t^{i,N_c}]^T$, $\chi_t^{i,n} \in \{0, 1\}$ indicates whether satellite i allocates a beam to cell n in time slot t . Define $\mathbf{z} = [z_1^1, z_1^2, \dots, z_t^n, \dots, z_t^{N_c}]$, and $z_t^n = \sum_{i=1}^{N_s} \chi_t^{i,n}$ is also a binary variable indicating whether cell n is activated and served by any satellite's beam during time slot t .

As shown in Fig. 1, the data packet requests generated by users within each cell are uploaded to the satellite through the user link and stored in the onboard buffers. As discussed in Section 1, the traffic demand across cells exhibits significant spatial and temporal variability and encompasses multiple types. Without loss of generality, the packets are classified into two categories: emergency and normal, denoted by $c \in \{e, n\}$, respectively. Let $\rho_{t,c}^n$ represent the set of data packets of type c in cell n during

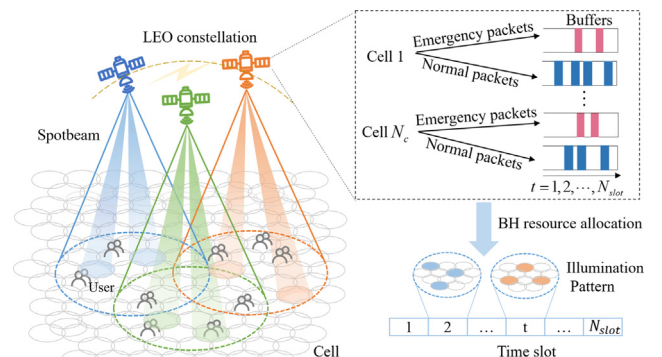


Fig. 1. Schematic diagram of multi-satellite coordinated beam-hopping scheduling system model.

time slot t . Based on the inherent properties of the data packets and the constraints imposed by the onboard buffers, packets are discarded if they exceed their Time-To-Live (TTL) without being responded or the queue length exceeds the maximum allowable limit. According to the waiting time, we further subdivide $\rho_{t,c}^n$ into $\rho_{t,c}^n = \{\phi_t^{n,1,c}, \phi_t^{n,2,c}, \dots, \phi_t^{n,l,c}, \dots, \phi_t^{n,\text{TTL},c}\}$, where $\phi_t^{n,l,c}$ denotes the number of type- c packets in cell n that have been queued for l time slots. Thus, the average waiting delay of type- c packets in cell n at time slot t can be derived as

$$\tau_t^{n,c} = \frac{\sum_{l=1}^{\text{TTL}_c} l \cdot \phi_t^{n,l,c}}{\sum_{l=1}^{\text{TTL}_c} \phi_t^{n,l,c}}, c = \{e, n\}. \quad (1)$$

For modeling convenience, all user requests within a given cell are aggregated into a single virtual user node located at the cell center. When the cell is served by a satellite beam, this virtual user, equipped with a receiving antenna, establishes a communication link with the beam. In the context of LEO satellite communication, the channel is inherently complex, involving both large-scale losses such as atmospheric attenuation, ionospheric effects, and free-space path loss, as well as small-scale fading due to factors like multipath propagation and doppler shift. Therefore, we referenced the description of LEO channel characteristics models outlined in 3GPP TR 38.821 (3GPP, 2021) and the relevant attenuation models defined by International Telecommunication Union (ITU), which provide a more robust and realistic representation of the propagation environment in NTN. Specifically, we adopted the Rice fading model to comprehensively account for both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) scenarios. The channel gain between the j -th beam of satellite i and cell n is denoted as

$$h_n^{i,j} = g_n^{i,j} \sqrt{\frac{k_n^{i,j}}{1 + k_n^{i,j}}} + \tilde{g}_n^{i,j} \sqrt{\frac{1}{1 + k_n^{i,j}}}, \quad (2)$$

where $k_n^{i,j}$ is the Rician factor, $g_n^{i,j}$ and $\tilde{g}_n^{i,j}$ represent the deterministic LoS and the NLoS component, respectively. Specifically, $g_n^{i,j} = \gamma_a \sqrt{G_{i,j}^T G_n^R \lambda / (4\pi d_{i,n})}$, where $G_{i,j}^T$ and G_n^R respectively represent the transmitting and receiving antenna gains, $d_{i,n}$ denotes the distance between satellite i and cell n , λ is the wavelength. γ_a is the attenuation factor, which is the result of a comprehensive calculation of atmospheric and ionospheric attenuation as specified in the ITU documents (ITU-R, 2017; ITU-R, 2019). The NLoS part satisfies $\tilde{g}_n^{i,j} \sim \mathcal{CN}(0, |g_n^{i,j}|^2)$.

$G_{i,j}^T$ and G_n^R are determined respectively by antenna radiation pattern of the satellite and the user. In this paper, the satellite's transmitting antenna radiation pattern is mod-

eled in accordance with the ITU-R S.1528 recommendation (ITU-R, 2001), which provides a standardized framework for multi-beam antennas and rigorously characterizes the main lobe and side lobe regions. The radiation pattern is given by

$$G^T(\theta) = \begin{cases} G_{\max}^T, 0 \leq \theta \leq \theta_b, \\ G_{\max}^T - 3(\theta/\theta_b)^2, \theta_b < \theta \leq Y, \\ G_{\max}^T + L_s - 25 \log(\theta/Y), Y < \theta \leq Z, \\ L_F, Z < \theta \leq 180, \end{cases} \quad (3)$$

where θ is the off-axis angle in degrees and $G^T(\theta)$ is the gain at angle θ from the main beam direction (dBi). $G_{\max}^T = 10 * \log_{10}(\eta \cdot (180/\theta_b)^2)$ is the maximum gain in the main lobe, in which θ_b is the 3 dB beamwidth and η is the antenna efficiency. $L_s = -6.75(\text{dB})$ represents the main beam and near-in side-lobe mask cross point below peak gain. $L_F = 0(\text{dBi})$ denotes the far-out side-lobe level. $Y = 1.5\theta_b$ and $Z = Y \times 10^{0.04(G_m + L_s - L_F)}$.

In addition, the user's receiving antenna radiation pattern is modeled in accordance with ITU-R S.465–6 (ITU-R, 1990), which specifies the gain attenuation characteristics of the main lobe, side lobe, and back lobe and is especially suitable for interference coordination in GSO/NGSO coexistence scenarios. Specifically, the radiation pattern has the form as

$$G^R(\theta) = \begin{cases} G_{\max}^R, 0 \leq \theta < \theta_{\min}, \\ 32 - 25 \log \theta, \theta_{\min} \leq \theta < 48^\circ, \\ -10, 48^\circ \leq \theta < 180^\circ, \end{cases} \quad (4)$$

in which θ is off-axis angle in degrees and $G^R(\theta)$ is the gain at angle θ from the main beam direction (dBi). $G_{\max}^R = 10 * \log_{10}(\frac{\pi D}{\lambda})^2$ denotes the maximum gain in the main lobe, where D is the diameter of the user antenna's aperture and λ is the wavelength. $\theta_{\min} = \max(1^\circ, 100\lambda/D)$ for $D/\lambda \geq 50$ and $\theta_{\min} = \max(2^\circ, 114(D/\lambda)^{-1.09})$ for $D/\lambda < 50$.

Then the signal-to-interference-plus-noise ratio (SINR) is derived as

$$\text{SINR}_i^{n,j} = \frac{|h_n^{i,j}|^2 p_i^{i,j}}{N + \sum_{j' \in \mathcal{J} \setminus \{j\}} |h_n^{i,j'}|^2 p_i^{i,j'} + \sum_{j' \in \mathcal{I}_s \setminus \{i\}} |h_n^{i',j'}|^2 p_i^{i',j'}}, \quad (5)$$

in which $p_i^{i,j}$ represents the power allocated for the i -th satellite's j -th beam. $N = KT_{\text{noise}}B_i^{i,j}$ stands for the noise power, which is calculated from equivalent system noise temperature T_{noise} , Boltzmann's constant K and bandwidth $B_i^{i,j}$. The last two terms in the denominator represent the CCI caused by other beams within the same satellite and by beams from other satellites.

To mitigate interference, the system abandons full-frequency reuse and instead employs adaptive beam illumination patterns in conjunction with flexible bandwidth

allocation. Specifically, the total available bandwidth B_{tot} is evenly divided into F orthogonal subbands, denoted by the set $\mathcal{F} = \{f | f = 1, 2, \dots, F\}$. Each active beam is assigned a contiguous block of subbands, and the binary indicator variable $a_t^{i,j,f} \in \{0, 1\}$ denotes whether beam j of satellite i occupies subband f at time slot t . If a beam is allocated $w_t^{i,j}$ subbands, its corresponding bandwidth can be calculated as $B_t^{i,j} = w_t^{i,j} \cdot B_0$, where $B_0 = B_{\text{tot}}/F$ denotes the bandwidth of a single subband. To avoid CCI, beams with interference relationships must be allocated mutually exclusive subband sets.

According to Shannon's theorem, the channel capacity of cell n in time slot t can be expressed as

$$C_t^n = z_t^n \cdot B_t^{i,j} \cdot \log_2(1 + \text{SINR}_t^{nj}). \quad (6)$$

The total traffic demand in cell n during time slot t is denoted as $d_t^n = \rho_{t,e}^n + \rho_{t,n}^n$, then the throughput of the cell in time slot t is denoted as

$$\Omega_t^n = \min(d_t^n, C_t^n). \quad (7)$$

2.2. Optimization problem formulation

In the BH satellite communication system, multiple satellite beams dynamically illuminate ground cells at appropriate time slots to accommodate traffic demands that vary across both space and time. The scheduling strategy aims to achieve two primary objectives: (1) minimizing inter-satellite load imbalance to prevent resource overload or underutilization, and (2) maximizing system throughput to accommodate more service requests. Furthermore, emergency services, such as disaster response communications, mandate a 100% access guarantee with minimal response latency. This requirement is addressed via a priority-based scheduling mechanism, rather than being explicitly incorporated into the optimization objectives. Based on these considerations, the overall BH resource scheduling problem is formulated as a constrained multi-objective optimization problem in (8).

$$\mathcal{P}_0 : \min_{\{u_n^i\}} \frac{1}{N_s} \sum_{i=1}^{N_s} \left(L_i - \frac{1}{N_s} \sum_{i=1}^{N_s} L_i \right)^2 \quad (8a)$$

$$\max_{\{z_t, a_t^{i,j,f}, p_t^{i,j}\}} \sum_{t=1}^T \sum_{n=1}^{N_c} \Omega_t^n \quad (8b)$$

$$\text{s.t. } C_1 : u_n^i \leq \xi_n^i, \quad \forall n, i \quad (8c)$$

$$C_2 : \sum_{i=1}^{N_s} u_n^i = 1, \quad \forall n \in \mathcal{N}_{\text{vis}} \quad (8d)$$

$$C_3 : L_i = \sum_{n=1}^{N_c} u_n^i d_n, \quad \forall i \quad (8e)$$

$$C_4 : \sum_{n=1}^{N_c} \chi_t^{i,n} \leq J, \quad \forall i, t \quad (8f)$$

$$C_5 : s_t^{i,j}, w_t^{i,j} \in \mathcal{F}, \quad s_t^{i,j} + w_t^{i,j} - 1 \leq F, \quad \forall i, j, t \quad (8g)$$

$$C_6 : a_t^{i,j,f} = 1 \iff s_t^{i,j} \leq f \leq s_t^{i,j} + w_t^{i,j} - 1, \quad \forall i, j, t, f \quad (8h)$$

$$C_7 : a_t^{i,j,f} + a_t^{k,l,f} \leq 1, \quad \forall ((i, j), (k, l)) \in \mathcal{I}_t, \quad (8i)$$

$$C_8 : \sum_{j=1}^J p_t^{i,j} \leq P_{\text{sat}}, \quad \forall i, t \quad (8j)$$

$$C_9 : 0 \leq p_t^{i,j} \leq P_{\text{max}}, \quad \forall i, j, t \quad (8k)$$

$$C_{10} : u_n^i, \xi_n^i, \chi_t^{i,n}, a_t^{i,j,f} \in \{0, 1\}, \quad \forall i, n, j, t, f \quad (8l)$$

where $\{u_n^i, \chi_t^{i,n}, p_t^{i,j}, a_t^{i,j,f}\}$ are the decision variables to be optimized. L_i denotes the load of satellite i , which is the sum of the traffic of all cells assigned to it as defined in (8e). (8c) and (8d) jointly enforce that every cell within the visible set $\mathcal{N}_{\text{vis}} = \{n | \sum_i \xi_n^i \geq 1\}$ is associated with exactly one satellite. Here, the binary variable $\xi_n^i \in \{0, 1\}$ indicates the visibility of satellite i to cell n (equal to 1 if visible, and 0 otherwise), and u_n^i denotes the association decision between cell n and satellite i . (8f) constrains that each satellite can generate up to J beams simultaneously. Constraints (8g)-(8h) enforce that each active beam occupies a set of contiguous subbands. Specifically, $s_t^{i,j}$ denotes the starting index of the allocated subband block, and $w_t^{i,j}$ indicates the number of subbands assigned to beam j of satellite i , which must not exceed the total number F of available subbands. (8i) eliminates CCI by ensuring non-overlapping frequency allocation for all interfering beam pairs $((i, j), (k, l)) \in \mathcal{I}$. The interference set $\mathcal{I}$, to be detailed in the following section, captures both intra-satellite ($i = k$) and inter-satellite ($i \neq k$) beam interference. (8j) and (8k) regulate the transmission power of satellite beams, where P_{sat} denotes the total available power and P_{max} is the maximum allowable transmit power per beam.

As formulated above, the proposed BH scheduling problem is a multi-objective mixed-integer nonlinear programming (MINLP) problem. The presence of both discrete and continuous variables significantly increases the problem's complexity. Nonlinear objectives involving logarithmic throughput and load balancing, combined with interference-avoidance constraints further contribute to the problem's non-convexity and render $\mathcal{P}_0$ NP-hard (Lin et al., 2023). Consequently, it's intractable to obtain the global optimum within acceptable complexity. We therefore decompose the problem into three tractable subproblems and develop low-complexity solutions accordingly in the next section.

3. Problem decomposition and solving method

To render the complex $\mathcal{P}_0$ tractable, the original problem is decomposed into three coordinated subproblems, namely satellite-cell association problem, BH illumination pattern and bandwidth joint allocation problem, and power optimization problem. First, leveraging the multi-

coverage characteristic of NGSO constellations, we formulate the satellite–cell association problem to determine the serving satellite for each cell over the scheduling period by optimizing binary variables u_n^i . This step establishes quasi-static serving relationships, balancing inter-satellite load and ensuring operational stability. Subsequently, conditioned on the association results u_n^i , the BH illumination pattern χ_t is optimized per time slot, followed by frequency subband allocation $a_t^{i,j,f}$ based on the interference graph of activated beams. This sequential allocation strategy is designed to mitigate inter-beam interference explicitly while adapting to dynamic traffic. Finally, with illumination pattern χ_t and subband assignments $a_t^{i,j,f}$ fixed, the power allocation problem is solved to determine the transmit power $p_t^{i,j}$ for each active beam, thereby fine-tuning resource usage to match actual demand. As illustrated in Fig. 2, this structured approach ensures computational feasibility while effectively balancing load, enhancing throughput, and suppressing interference.

3.1. Satellite-cell association problem and solution

Due to the highly uneven distribution of traffic demand across ground cells, conventional association strategies often cause load imbalance by concentrating traffic on a few satellites. Leveraging the inherent multi-satellite coverage capability of NGSO constellations, we formulate an association strategy that minimizes inter-satellite load disparity as follows,

$$\mathcal{P}_1 : (8a) \quad (9a)$$

$$\text{s.t. } (8c) - (8e), (8l). \quad (9b)$$

$\mathcal{P}_1$ is a classical 0–1 quadratic integer programming (QIP) problem, and similar formulations for user-satellite association have been studied in previous works (Jiang et al., 2022; Pachler et al., 2024). However, our scenario specifically focuses on determining the serving satellite for a scheduling period, which differs in both objective and operational constraints from the long-term stable access schemes in those works, thus requiring a new efficient solution. While exact methods such as branch-and-bound or convex relaxation offer opti-

malty guarantees, their prohibitive complexity limits practical efficiency, especially for large-scale instances. As a result, heuristic approaches become popular for achieving near-optimal solutions within acceptable timeframes. A prevalent method is the greedy strategy (Lin et al., 2023) that prioritizes cells by descending traffic demand and assigns each cell sequentially to the currently least-loaded visible satellite. Although easy to implement and effective in some scenarios, this approach is sensitive to the initial sorting and lacks a backtracking mechanism, making it prone to suboptimal allocations.

To overcome the above-mentioned drawbacks without incurring significant additional complexity, we propose a simulated annealing (SA) based algorithm that leverages the greedy heuristic as its initial solution and integrates probabilistic reassignment, thereby enabling a more exhaustive exploration of the solution space and avoiding entrapment in suboptimal local minima. Below we detail the proposed SA algorithm for $\mathcal{P}_1$, including a narrative description and key equations.

Firstly, the initialization is performed based on the greedy strategy: sort all cells n in descending order of demand d_n and assign each cell to the least-loaded satellite among its visible set $\Xi_n = \{i \in \mathcal{N}_s | \zeta_n^i = 1\}$. Denote the resulting binary assignment matrix as $\mathbf{U}^{(0)} = [\mathbf{u}_n^{i(0)}] \in \mathbb{B}^{N_c \times N_s}$. At each iteration, starting from the current assignment $\mathbf{U}$, we generate a neighboring solution $\mathbf{U}'$ by selecting one cell n and reassigning it to another satellite i' with $i' \in \Xi_n$. To enhance the convergence efficiency of the SA algorithm, we propose a multi-phase biased neighborhood generation mechanism that dynamically screen traffic-sensitive cells to guide local optimization while preserving global exploration capabilities. Specifically, the procedure for generating neighboring solutions is structured as follows.

A candidate cell n is selected through a hierarchical probabilistic strategy. Firstly, with probability p_1 , a cell is randomly chosen from $\mathcal{N}_c$. Otherwise, an overloaded satellite $i \in \mathcal{N}_s$ with $L_i > \bar{L}$ is selected, and the set of cells currently associated with satellite i is denoted by $\Theta_i = \{n | u_n^i = 1\}$. Then, based on the inter-satellite load gap $\Delta L = \max(\mathbf{L}) - \min(\mathbf{L})$, a cell $n \in \Theta_i$ satisfying the condition $d_n \leq \Delta L$ is prioritized with probability p_2 . Finally, the reassignment of the chosen cell n to a new satellite is performed under visibility constraints and load-balancing considerations. With probability p_3 , cell n is reassigned to a visible satellite $i' \in \Xi_n$ with $L_{i'} < \bar{L}$; otherwise, it's randomly reallocated within the visible set Ξ_n to maintain the global exploration capabilities of the algorithm.

We measure solution quality by the load-variance energy as

$$E(\mathbf{U}) = \frac{1}{N_s} \sum_{i=1}^{N_s} (L_i(\mathbf{U}) - \bar{L}(\mathbf{U}))^2, \quad (10)$$

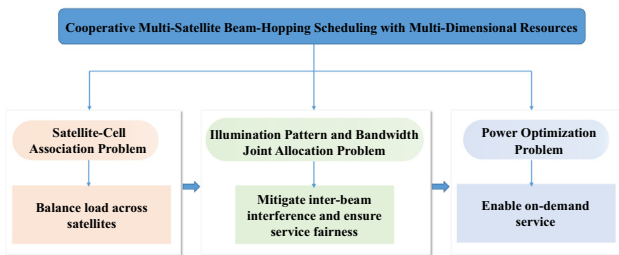


Fig. 2. Problem decomposition for cooperative multi-satellite beam-hopping.

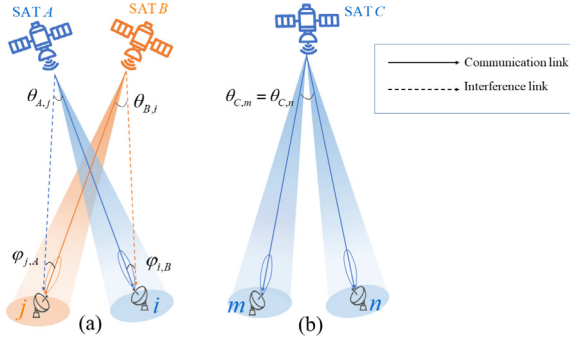


Fig. 3. Illustration of co-channel interference between beams from (a) different satellites and (b) the same satellite.

where $L_i(\mathbf{U}) = \sum_n d_n u_n^i$ and $\bar{L}(\mathbf{U}) = \frac{1}{N_s} \sum_i L_i$. Then the energy change can be calculated by $\Delta E = E(\mathbf{U}') - E(\mathbf{U})$. If $\Delta(E) \leq 0$, the move is unconditionally accepted; otherwise it is accepted with probability $e^{-\Delta E/T}$, where T is the current temperature. After the acceptance decision, the temperature is reduced according to a geometric cooling schedule $T \leftarrow \alpha T$ with $0 < \alpha < 1$. This process—neighbor generation, energy evaluation, probabilistic acceptance, and temperature update—repeats until either T falls below a prescribed minimum $T_{\min}$ or the maximum number of iterations is reached. The overall process of the algorithm is summarized in Algorithm 1.

Algorithm 1. Simulated Annealing Solution for Satellite-Cell Association Problem

Input: Traffic demand of cells d_n , the number of satellites N_s , the number of cells N_c , visibility relationship between satellites and cells ξ_n^i , the initial temperature T_0 , cooling rate α , minimum temperature $T_{\min}$, max iterations $I_{\max}$

Initialize:

Generate initial allocation $\mathbf{U}^{(0)}$ according to greedy strategy
 Calculate load-variance energy
 $E^{(0)} = \frac{1}{N_s} \sum_{i=1}^{N_s} (L_i - \bar{L})^2$
 Set $T = T_0, k = 0, E^* = E^{(0)}, \mathbf{U}^* = \mathbf{U}^{(0)}$
 1: **while** $k < I_{\max}$ **and** $T > T_{\min}$ **do**
 2: **if** $\text{rand}(0,1) < p_1$ **then**
 3: Randomly select a satellite i and a cell n such that $u_n^i = 1$
 4: **else**
 5: Select an overload satellite $i \in \mathcal{N}_s$ such that $L_i > \bar{L}$
 6: Get the current serving cell set of satellite $i : \Theta_i = \{n | u_n^i = 1\}$
 7: Calculate the maximum inter-satellite load difference:
 $\Delta L = \max(\mathbf{L}) - \min(\mathbf{L})$

8: **if** $\text{rand}(0,1) < p_2$ **then**
 9: Select a cell $n \in \Theta_i$ satisfying $d_n \leq \Delta L$
 10: **else**
 11: Randomly select a cell $n \in \Theta_i$
 12: **end if**
 13: **end if**
 14: Get the set of satellites visible to cell $n : \Xi_n = \{i | \xi_n^i = 1\}$
 15: **if** $\text{rand}(0,1) < p_3$ **then**
 16: Select satellite $i' \in \Xi_n$ that satisfies $L_{i'} < \bar{L}$
 17: **else**
 18: Reallocate cell n to any satellite i' in Ξ_n
 19: **end if**
 20: Obtain neighbor solution $\mathbf{U}'^{(k)}$ by setting $u_n^i = 0, u_n^{i'} = 1$
 21: Calculate $\Delta E = E(\mathbf{U}'^{(k)}) - E(\mathbf{U}^{(k)})$ according to Eq. (10)
 22: **if** $\Delta E \leq 0$ or $\text{rand}(0,1) < e^{-\Delta E/T}$ **then**
 23: $\mathbf{U}^{(k)} = \mathbf{U}'^{(k)}, E^{(k)} = E(\mathbf{U}'^{(k)})$
 24: Update the load of each satellite:
 $L_i = L_i - d_n, L_{i'} = L_{i'} + d_n$
 25: **if** $E^{(k)} < E^*$ **then**
 26: $\mathbf{U}^* = \mathbf{U}^{(k)}, E^* = E^{(k)}$
 27: **end if**
 28: **end if**
 29: $T = \alpha \cdot T, k = k + 1$
 30: **end while**
Output: Optimal allocation solution $\mathbf{U}^*$

3.2. BH illumination pattern and bandwidth joint allocation problem and solution

Given the established satellite-cell association of the BH scheduling period, the subsequent task is to determine the BH illumination pattern of each satellite at each time slot. The objective is to dynamically satisfy the spatiotemporally heterogeneous traffic demands of cells as much as possible, thereby enhancing overall system capacity. However, simultaneous activation of multiple beams across the constellation may introduce severe co-channel interference (CCI). To address this, frequency subbands must be flexibly allocated to the active beams based on their interference relationships. Drawing on the concept of interference sets from prior work (Pachler et al., 2024), we formulate the problem using a graph model to obtain a solution.

Based on the satellite-cell association results, potential interference scenarios are analyzed as illustrated in Fig. 3. Interference may occur when geographically adjacent cells are simultaneously served by beams from different satellites (Fig. 3(a)) or the same satellite (Fig. 3(b)), due to side-lobe signal leakage at both the transmitting and receiving antennas. In Fig. 3(a), cell i is served by satellite A , while the

adjacent cell j is served by satellite B during the BH scheduling period. $\theta_{A,j}$ and $\theta_{B,i}$ denote the off-axis deviations of the interfering satellite beams from the main lobe direction toward the unintended target cells, while $\varphi_{j,A}$ and $\varphi_{i,B}$ represent the angles of the receiving antenna deviating from the main axis. Interference is considered significant when both $\theta \leq \theta_{th}$ and $\varphi \leq \varphi_{th}$, where θ_{th} and φ_{th} are the preset interference angle thresholds. Similarly, in Fig. 3 (b), adjacent cells m and n are simultaneously served by two beams from the same satellite C . In this scenario, the communication links from satellite C to cell m and cell n inherently act as mutual interfering links, and severe interference is more likely to occur because both receiving antennas are precisely steered to the satellite. Notably, the off-axis angles satisfy $\theta_{C,m} = \theta_{C,n}$ and interference is deemed severe if $\theta \leq \theta_{th}$.

An interference graph can be constructed based on the aforementioned interference analysis as shown in Fig. 4. In this graph, each cell is represented as a vertex, and an edge is drawn between two vertices if the corresponding cells exhibit potential CCI. According to graph theory, the degree of a vertex is defined as the number of edges connected to it. As previously discussed, cells with zero degree can utilize the full available bandwidth, whereas those with nonzero degrees require flexible subband allocation to mitigate interference. The interference set $\mathcal{I}$ illustrated in Fig. 4 consists of five interference pairs in total.

When determining the illumination pattern, two key objectives are considered: (1) maximizing the throughput, and (2) ensuring inter-cell fairness to prevent persistent neglect of low-demand cells. To quantify fairness, we introduce the concept of cumulative satisfaction rate and define the cumulative satisfaction rate of cell n at time slot t as μ_t^n . It's worth noting that the demand d_t^n encompasses not only newly generated requests at slot t but also incorporates unfulfilled demand from preceding time slots. Denote the expected service capacity of cell n at slot t as

$$\tilde{R}_t^n = B_t^{i,j} \cdot \log_2 \left(1 + \frac{|h_n^{i,j}|^2 P_{\max}}{n_0 B_t^{i,j} + \sum_{(k,l) \in \mathcal{I}_t^{i,j}} |h_n^{k,l}|^2 P_{\max}} \right), \quad (11)$$

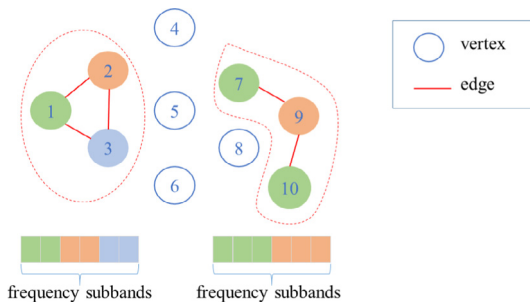


Fig. 4. An example of the interference graph with 10 cells.

where $n_0 = KT_{noise}$ is the noise power spectral density, $\mathcal{I}_t$ denotes the set of all interfering beam pairs and $\mathcal{I}_t^{i,j} = \{(k,l) \mid ((i,j), (k,l)) \in \mathcal{I}_t\}$ represents the set of all beams that interfere with (i,j) . Accordingly, the cumulative satisfaction ratio at slot t can be recursively computed as Eq. (12).

$$\mu_t^n = \frac{\sum_{k=1}^{t-1} \left(d_k^n \cdot \prod_{j=k}^{t-1} \mu_j^n \right) + \tilde{R}_t^n}{\sum_{k=1}^{t-1} \left(d_k^n \cdot \prod_{j=k}^{t-1} \mu_j^n \right) + d_t^n}, \quad t \geq 2. \quad (12)$$

Building upon the above discussion, the BH illumination pattern and bandwidth allocation problem can be jointly formulated as

$$\mathcal{P}_2 : \max_{\mathbf{z}_t, \mathbf{a}_t^{i,j}} \sum_{n=1}^{N_c} \min \left(\tilde{R}_t^n, d_t^n \right) \quad (13a)$$

$$\max \min_{\mathbf{z}_t, \mathbf{a}_t^{i,j}} \tilde{\mu}_t^n \quad (13b)$$

$$\text{s.t. } (8f) - (8i), (11), (12). \quad (13c)$$

Given the severe variable coupling observed in $\mathcal{P}_2$, we propose a simplified low-complexity algorithm to obtain acceptable suboptimal solutions. Specifically, to ensure fairness, we first calculate the cumulative satisfaction rate of each cell over a sliding window of W previous time slots. Underserved cells with satisfaction rates below a predefined threshold are prioritized for scheduling. Subsequently, under the per-satellite beam constraint, an integer linear programming (ILP) problem is then solved to select a set of cells with the highest total demand, forming the illumination pattern. Notably, due to the higher priority of emergency traffic, this process is first applied to the set of emergency cells, followed by the remaining regular cells with updated constraints. Finally, for all activated cells, a subband allocation is performed based on the corresponding interference graph. As an illustrative example in Fig. 4, assume the total bandwidth is partitioned into six equal-width subbands. In the corresponding interference graph, cells 1, 2, and 3 form a cycle (i.e., a 3-clique), implying mutual interference among them. According to graph coloring theory, at least three orthogonal subband groups are required to avoid conflicts, resulting in each cell being assigned two subbands. In contrast, cells 7, 9, and 10 constitute a linear chain structure, which only requires two colors to achieve interference-free allocation, thereby allowing each cell to occupy three subbands. The complete solution procedure is summarized in Algorithm 2.

Algorithm 2. Graph Based Solution for Illumination Pattern and Bandwidth Joint Allocation Problem

Input: Traffic demand of emergency and normal types of cells $d_t^n = [\rho_{t,e}^n, \rho_{t,n}^n]$, satellite-cell association result $\mathbf{U} = [u_n^i]$, interference angle threshold $\theta_{th}, \varphi_{th}$, number of subbands F , beam number limit J , cumulative satisfaction rate μ_t^n , satisfaction rate threshold μ_{th} , demand threshold d_0

Initialize:

- Set interference set $\mathcal{I} = \emptyset$ and $\mathbf{z}_t = \mathbf{0}$
- 1: Calculate the off-axis angles θ, φ based on the satellite and cell location coordinates
- 2: **if** $\theta \leq \theta_{th}$ and $\varphi \leq \varphi_{th}$ **then**
- 3: Add the edge to the interference graph $\mathcal{G}$ and add interference pairs $(i, j), (k, l)$ to $\mathcal{I}$
- 4: **end if**
- 5: Define the underserved cell set $\Lambda = \{n : \mu_t^n < \mu_{th}, d_t^n > d_0\}$
- 6: **for** $i = 1$ **to** N_s **do**
- 7: Traverse $n \in \Lambda$ in descending order of d_n , if $u_n^i = 1$ and $\sum_{n=1}^{N_c} \chi_t^{i,n} < J$, set $\chi_t^{i,n} = 1$
- 8: **end for**
- 9: Solve ILP to maximize emergency traffic $\sum \chi_t^{i,n} \rho_{t,e}^n$ under (8f)
- 10: Solve ILP to maximize normal traffic $\sum \chi_t^{i,n} \rho_{t,n}^n$ under (8f)
- 11: Obtain all active cells and $\mathbf{z}_t$
- 12: Extract subgraphs of active cells from $\mathcal{G}$ and get interfering clusters $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$
- 13: **for** each cluster $\mathcal{C}_k$ **do**
- 14: Compute minimum number of colors via graph coloring
- 15: **for** each cell $n \in \mathcal{C}_k$ **do**
- 16: Allocate continuous sub-bands $\{f\}$ based on the coloring result
- 17: Set $a_t^{i,j,f} = 1$ if beam (i, j) serves cell n on subband f
- 18: **end for**
- 19: **end for**
- Output:** Illumination pattern $\mathbf{z}_t$ and bandwidth allocation $a_t^{i,j,f}$ of each beam

3.3. Bandwidth-adaptive power allocation problem and solution

In this subsection, the power of each active beam will further be optimized under previously determined BH illumination pattern and bandwidth allocation results. Since inter-satellite interference has already been addressed in the preceding subproblem, the optimization can now be performed independently within each satellite. For each satellite, the objective is to maximize the sum throughput of all active beams, which can be formulated as follows,

$$\mathcal{P}_3 : \max_{p_t^{i,j}} \sum_{n=1}^{N_c} \hat{\Omega}_t^n \quad (14a)$$

$$\text{s.t. } \hat{\Omega}_t^n = \min \{d_t^n, \hat{C}_t^n\} \quad (14b)$$

$$\hat{C}_t^n = z_t^n \cdot B_t^{i,j} \cdot \log_2 \left(1 + \frac{|h_n^{i,j}|^2 p_t^{i,j}}{n_0 B_t^{i,j}} \right) \quad (14c)$$

$$(8j) - (8k), \quad (14d)$$

where $B_t^{i,j}$ and $z_t^n = \sum_{i=1}^{N_s} \chi_t^{i,n}$ are both known quantities.

$\mathcal{P}_3$ is convex and can be efficiently solved via the standard Lagrangian dual method, as established in prior works (Hong et al., 2008). We adapt this method to our beam-hopping context by introducing a bandwidth-aware water-filling process, in which the water level is jointly determined by the pre-allocated subband and channel conditions, and further incorporate a demand-aware compensation step to improve practical efficiency. In our scenario, an illuminated cell (n) and a serving beam (j) are synonymous concepts, so we rewrite z_t^n as z_t^j to simplify notation and avoid confusion. The Lagrangian function can be constructed as

$$\mathcal{L} = \sum_{j=1}^J z_t^j \cdot B_t^{i,j} \cdot \log_2 \left(1 + \frac{|h_n^{i,j}|^2 p_t^{i,j}}{n_0 B_t^{i,j}} \right) - \lambda \left(\sum_{j=1}^J p_t^{i,j} - P_{sat} \right), \quad (15)$$

where $\lambda \geq 0$ is the Lagrange multiplier associated with the total power constraint. To obtain the optimal solution, we take the derivative of $\mathcal{L}$ with respect to $p_t^{i,j}$ and set it to zero as follows,

$$\frac{\partial \mathcal{L}}{\partial p_t^{i,j}} = \frac{z_t^j \cdot B_t^{i,j} |h_n^{i,j}|^2}{\ln 2 \cdot (n_0 B_t^{i,j} + |h_n^{i,j}|^2 p_t^{i,j})} - \lambda = 0. \quad (16)$$

Then the optimal power allocation can be derived as

$$p_t^{i,j} = \left[\frac{z_t^j B_t^{i,j}}{\lambda \ln 2} - \frac{n_0 B_t^{i,j}}{|h_n^{i,j}|^2} \right]^+, \quad (17)$$

where $[x]^+ = \max(x, 0)$ ensures non-negative constraint. Eq. (17) is a classic water-filling solution, where the water level $1/(\lambda \ln 2)$ must be adjusted such that the total power constraint (8j) is satisfied.

The optimal λ can be computed with high precision via bisection search. Let λ_{min} and λ_{max} denote the initial lower and upper search bounds, respectively. In each iteration, a tentative value $\lambda = (\lambda_{min} + \lambda_{max})/2$ is selected, and the corresponding power allocation $p_t^{i,j}$ is computed according to Eq. (17). The total transmit power $P_{sum} = \sum_{j=1}^J p_t^{i,j}$ is then compared with total available power P_{sat} . Specifically, if the total power exceeds P_{sat} , which indicates that λ is too small and should be increased, the lower bound λ_{min} is updated to the current λ . Conversely, if $P_{sum} < P_{sat}$, the upper bound λ_{max} should be updated to the current λ . The search terminates when $P_{sat} - P_{sum} < \epsilon$, in which

$\epsilon > 0$ is a predefined threshold. Due to the convexity of the optimization problem, this procedure guarantees a globally optimal power allocation.

It is worth noting that the above water-filling solution provides an optimal capacity-oriented power allocation, which solely maximizes the aggregated beam capacity under different bandwidth. However, achievable throughput is also constrained by actual traffic demand as depicted in Eq. (14b). Allocating excess power to low-demand beams wastes resources and reduces power for more demanding beams. To address this inefficiency, a demand-aware compensation mechanism is introduced. For beams whose capacity exceeds their traffic demand, the minimum required transmit power that just satisfies the demand can be recalculated under the given bandwidth and channel gain. The redundant power is reclaimed and then redistributed to beams with unsatisfied demand in proportion to their unmet demand and channel conditions. Moreover, the per-beam power limit $P_{\max}$ is strictly enforced during both initial allocation and redistribution. The complete power allocation algorithm is presented in Algorithm 3.

Algorithm 3. Power allocation with water-filling and demand-aware adjustment

Input: Illumination pattern $\chi_t^{i,n}$, cell demands d_t^n , bandwidth allocation $B_t^{i,j}$, channel gain $|h_n^{i,j}|^2$, total satellite power P_{sat} , max per-beam power $P_{\max}$, noise density n_0 , iteration cutoff condition $\epsilon = 1e - 6$

Initialize:

Set $P_{\text{sum}} = 0$ and give the initial lower and upper search bound $\lambda_{\min}, \lambda_{\max}, \lambda = (\lambda_{\min} + \lambda_{\max})/2$

- 1: **for** $i = 1$ to N_s **do**
- 2: Identify active beams $\mathcal{J}_i = \{j : \exists n, \chi_t^{i,n} = 1\}$
- 3: **while** $P_{\text{sat}} - P_{\text{sum}} > \epsilon$ **do**
- 4: **for** each beam $j \in \mathcal{J}_i$ **do**
- 5: Compute $p_t^{i,j}$ under current λ according to Eq. (17)
- 6: $p_t^{i,j} = \max(p_t^{i,j}, P_{\max})$
- 7: **end for**
- 8: Update $P_{\text{sum}} = \sum_{j \in \mathcal{J}_i} p_t^{i,j}$ and $\lambda_{\min}$ or $\lambda_{\max}$
- 9: $\lambda = (\lambda_{\min} + \lambda_{\max})/2$
- 10: **end while**
- 11: **for** each beam $j \in \mathcal{J}_i$ **do**
- 12: Compute capacity $\hat{C}_t^n$ according to Eq. (14c)
- 13: **if** $\hat{C}_t^n > d_t^n$ **then**
- 14: Compute minimal power p_{req} to satisfy d_t^n
- 15: Reclaim power:
 $P_{\text{sum}} = P_{\text{sum}} + (p_t^{i,j} - p_{\text{req}}), p_t^{i,j} = p_{\text{req}}$
- 16: **end if**
- 17: **end for**
- 18: **while** $P_{\text{sat}} - P_{\text{sum}} > \epsilon$ and any beam has unmet demand **do**
- 19: Sort beams with unmet demand in descending order of shortage

- 20: **for** each such beam j **do**
- 21: Recalculate power required P_{req}
- 22: $p_t^{i,j} = \min(P_{\max}, P_{\text{req}}, p_t^{i,j} + P_{\text{sat}} - P_{\text{sum}})$
- 23: Update P_{sum}
- 24: **end for**
- 25: **end while**
- 26: **end for**

Output: Power allocation $p_t^{i,j}$ of each beam

After sequentially solving the three subproblems, the complete BH scheduling solution can be obtained. The proposed framework operates through a hierarchical ground-satellite coordination mechanism with differentiated timescales. Leveraging the forecasted traffic demands from historical data and the visibility conditions from ephemeris information, the ground control center first solves the satellite-cell association problem at a coarse timescale of tens of seconds. The resulting association scheme or incremental update is then disseminated to the relevant satellites, incurring modest signaling overhead due to its infrequent updates. Subsequently, each satellite autonomously performs distributed beam illumination and bandwidth allocation, coordinating interference information with neighboring satellites via inter-satellite links. Finally, each satellite independently executes low-complexity, per-slot power allocation for its activated beams. This hierarchical decomposed approach ensures computational tractability, and a closed loop is formed as telemetry data is reported back to the ground for monitoring and long-term strategy refinement. The overall framework of proposed algorithm for addressing the original optimization problem $\mathcal{P}_0$ is summarized in Algorithm 4.

Algorithm 4. Overall algorithm framework to solve $\mathcal{P}_0$

Input: $N_c, N_s, d_t^n, \xi_n^i, T, T_0, \alpha, T_{\min}, I_{\max}, \theta_{\text{th}}, \varphi_{\text{th}}, W, F, J, \mu_{\text{th}}, d_0, h_n^{i,j}, P_{\text{sat}}, P_{\max}, n_0, \epsilon$

Initialize: $u_n^i = 0, \forall n, i, \mu_n^i = 0, \forall n$

- 1: Obtain satellite–cell association result $\{u_n^i\}$ using Algorithm 1
- 2: Construct interference graph $\mathcal{G}$ based on $\{u_n^i\}$ and angle thresholds
- 3: **for** $t = 1$ to T **do**
- 4: Obtain illumination pattern χ_t and bandwidth allocation $\{a_t^{i,j,f}\}$ using Algorithm 2
- 5: **for** $i = 1$ to N_s **do**
- 6: Obtain power allocation of active beams $p_t^{i,j}$ of satellite i using Algorithm 3
- 7: **end for**
- 8: Calculate Ω_t^n according to Eq. (7) and update demand d_t^n and cumulative satisfaction rate μ_t^n
- 9: **end for**

Output: Scheduling result $\{u_n^i, \chi_t, a_t^{i,j,f}, p_t^{i,j}\}$

Table 1
Simulation parameter.

Parameters	Values
System parameters	
Number of satellites N_s	1728
Number of satellite orbits	48
Orbit inclination	50
Number of satellites per orbit	36
Satellite altitude H	1145 km
Number of cells in target area N_c	92
Cell radius	44 km
Downlink frequency	18.7 GHz
Total onboard power P_{sat}	20 dBW
System bandwidth B_{tot}	250 MHz
Number of subbands F	6
Maximum beams for each satellite J	3
Maximum transmit power per beam P_{max}	17 dBW
3 dB beamwidth of satellite beams θ_b	3°
Maximum transmitting antenna gain G_{ik}^{max}	34.01 dBi
User antenna aperture diameter D	0.4 m
Maximum receiving antenna gain G_n^{max}	36.33 dBi
Minimum user elevation angle	40°
Interference angle thresholds θ, φ	10°, 25°
System noise temperature T_{noise}	338 K
Traffic rate of normal demand in cells $\rho_{i,n}^n$	60 Mbps ~ 2000 Mbps
Traffic rate of emergency demand in cells $\rho_{i,e}^n$	0 ~ 240 Mbps
Number of time slots in a scheduling period T	1000
Length of time slot T_{slot}	2 ms
TTL of emergency packets TTL_e	5 T_{slot}
TTL of normal packets TTL_n	40 T_{slot}
Algorithm parameters	
Initial temperature T_0	1000
Cooling rate α	0.95
Minimum temperature T_{min}	0.00001
Maximum iteration of SA algorithm I_{max}	500
Satisfaction rate threshold μ_{th}	30 %
Demand threshold d_0	100 Mbps

3.4. Complexity Analysis

This subsection analyzes the computational complexity of the proposed framework.

(1) Satellite–Cell Association (Algorithm 1): This step is performed only once at the beginning of each association window. It consists of a greedy initialization and SA refinement. The initialization costs $\mathcal{O}(N_c(\log N_c + N_{\text{max}}))$, where N_c is the number of ground cells and N_{max} is the maximum number of visible satellites per cell. The SA refinement incurs $\mathcal{O}(I_{\text{max}}(N_{\text{max}} + \log N_v))$ for I_{max} iterations, where N_v is the number of concurrently visible satellites for target area. The overall complexity of Algorithm 1 is therefore $\mathcal{O}(N_c(\log N_c + N_{\text{max}}) + I_{\text{max}}(N_{\text{max}} + \log N_v))$.

(2) BH Pattern Design and Subband Allocation (Algorithm 2): This algorithm is executed in each slot and comprises interference graph construction, ILP-based BH pattern design, active subgraph extraction and subband assignment. Since the graph remains unchanged during one scheduling epoch, its construction is a one-time cost

of $\mathcal{O}(E)$, where E denotes the number of edges. The complexity of the BH pattern determined based on ILP is $\mathcal{O}(N_c \log N_c)$. The active subgraph in a given slot contains at most JN_v vertices. In the worst case, every active cell interferes with all others, yielding $\mathcal{O}((JN_v)^2)$. Therefore, the overall per-slot complexity is $\mathcal{O}(E + T[N_c \log N_c + (JN_v)^2])$.

(3) Power allocation (Algorithm 3): The power allocation employs a water-filling algorithm solved via bisection. Let the number of bisection iterations required to achieve precision ε be $I_{\text{bis}} = \mathcal{O}(\log(1/\varepsilon))$. In each iteration, the power update for all active beams of a single satellite incurs $\mathcal{O}(J)$ operations. Thus, the overall cost is $\mathcal{O}(N_v J \log(1/\varepsilon))$. The subsequent demand-aware compensation requires $\mathcal{O}(N_v J \log J)$, which remains dominated by the bisection term. Therefore, the total complexity of the power allocation algorithm is $\mathcal{O}(N_v J \log(1/\varepsilon))$.

Therefore, the overall complexity is $\mathcal{O}(N_c(\log N_c + N_{\text{max}}) + I_{\text{max}}(N_{\text{max}} + \log N_v) + E + T[N_c \log N_c + (JN_v)^2 + N_v J \log(1/\varepsilon)])$ in each epoch. The complexity is dominated by terms related to the number of visible satellites N_v , which is geometry-bounded rather than scaling proportionally with the total constellation size. Although a quadratic term $(JN_v)^2$ exists, the number of active beams per satellite J is a small constant. Furthermore, the underlying interference graph is sparse, and the corresponding computation is amenable to parallelization. These factors collectively ensure the framework's favorable scalability for mega-constellations. Overall, the proposed framework prioritizes computational tractability, providing an integrated and efficient solution for multi-satellite BH scheduling.

4. Simulation results and performance analysis

4.1. Simulation parameter

In this paper, a Ka-band mega-constellation referred to GW (ITU, 2020) is simulated, where a total of 1728 LEO satellites are uniformly distributed across 48 orbital planes at an altitude of 1145 km with an inclination of 50 degrees. The target ground area includes 92 cells located in the Yangtze River Delta region of China, spanning a longitude range of [115°E, 123°E] and a latitude range of [27°N, 36°N]. The time slot length follows the DVB-S2X standard (ETSI, 2021), and other critical system parameters are based on the Starlink technical reports (LLC, 2018). The main system parameters are summarized in Table 1. To better capture the spatiotemporal heterogeneity of traffic demands and to evaluate the effectiveness of the proposed algorithm under realistic scenarios, the traffic volume in each cell is modeled based on real-world population density (WorldPop, 2023) and per capita GDP data.

It is important to clarify that the 2-s simulation duration, which consists of 1000 consecutive 2-ms time slots, is configured solely for performance evaluation and does

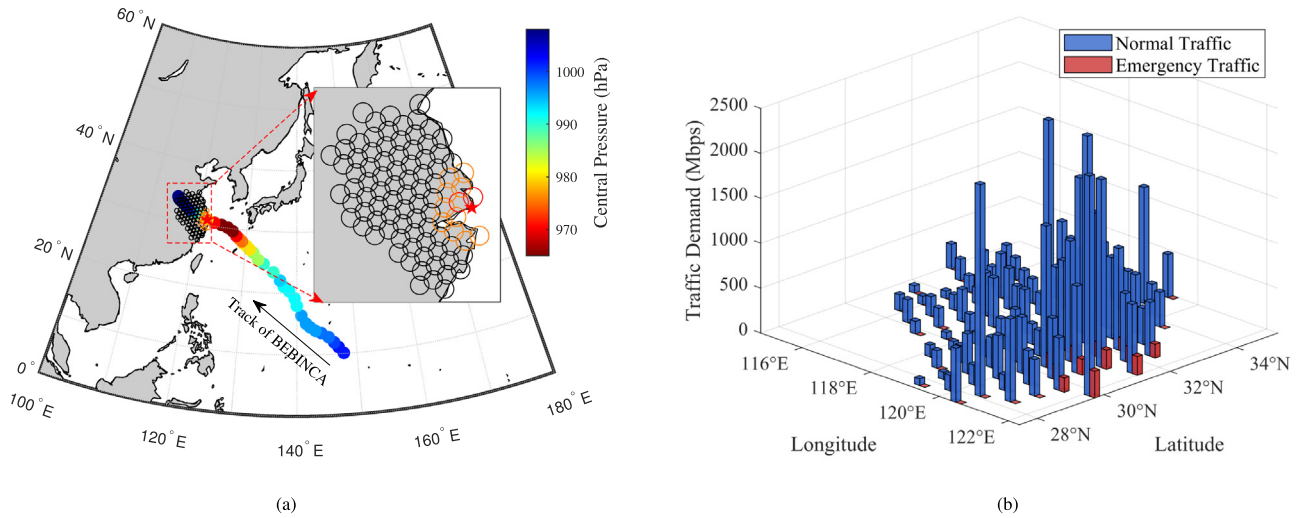


Fig. 5. Illustration of (a) simulation scenario and (b) spatial distribution of cell traffic demands.

not imply that the satellite–cell association must be recomputed every 2 s in practical deployments. As explained at the end of Section 3.3, the association algorithm operates at a much coarser timescale, for example with updates every 15 s. This distinction ensures that control signaling overhead remains manageable in real-world implementations.

Furthermore, considering the frequent typhoon activity in the Yangtze River Delta, we incorporate emergency communication traffic to represent real-time service demands. Specifically, we simulate emergency traffic demand based on the actual trajectory and wind speed profile of Typhoon BEBINCA (NII, 2024). The simulation scenario and spatial distributions of both normal and emergency traffic demands across the 92 cells are illustrated in Fig. 5.

Fig. 5a illustrates both the trajectory of Typhoon BEBINCA through the Yangtze River Delta and the distribution of the 92 target cells. Given its initial land-fall (UTC 2024/09/16 07:00) speed of 45 m/s, the typhoon can be considered quasi-static within the simulation period, as this velocity is negligible relative to the 44 km cell radius and the millisecond-level time-slot duration. The typhoon's center is marked with a red pentagram on the map. According to the typhoon center position and wind field parameters, the affected cells during the current scheduling period are color-coded based on the wind force level: cells affected within the radius of the Category 7 wind field are marked in orange, while those impacted by the Category 10 wind field are marked in red. Fig. 5b presents the distribution of both normal and emergency traffic demands across all cells. The traffic load is clearly highly uneven, with metropolitan areas such as Shanghai and Hangzhou exhibiting significantly higher demand due to their extremely high population densities and advanced economic development.

4.2. Performance analysis

4.2.1. Performance of satellite-cell association and load balancing

In order to balance the non-uniform traffic demand to each satellite, inter-satellite load balancing problem $\mathcal{P}_1$ is addressed using SA algorithm, as formulated in Section 3.1. We compare the proposed method with two baseline strategies:

- *MaxElevation*: assigns each cell to the satellite with the highest elevation angle during the current scheduling period.
- *MaxTime*: selects the satellite that provides the longest expected service duration for each cell.

Fig. 6 illustrates the satellite-cell association results under different schemes, where each pentagram denotes a visible satellite and the numeric labels within cells indicate their assigned satellite indices. It can be observed that 17 satellites are visible over the target area. Under the *MaxElevation* scheme, the cells are densely accessing only five satellites (SAT 2, 6, 11, 14 and 15), which will obviously increase the load of these satellites and result in highly skewed load distribution and inefficient resource utilization. The *MaxTime* scheme further exacerbates load imbalance by allocating all cells to only two satellites (SAT 5 and 13). In contrast, the proposed algorithm distributes the cell associations across all available satellites. Specifically, in Fig. 6c, some satellites (e.g., SAT 5 and 9) serve a large number of low-demand cells, while others (e.g., SAT 2 and 17) handle fewer but high-demand cells, thereby ensuring load balancing and enhancing overall system utilization.

Fig. 7 illustrates the convergence process of the proposed SA algorithm, in comparison with that of the standard SA. The convergence of the proposed SA features three distinct regimes: a rapid initial descent with approxi-

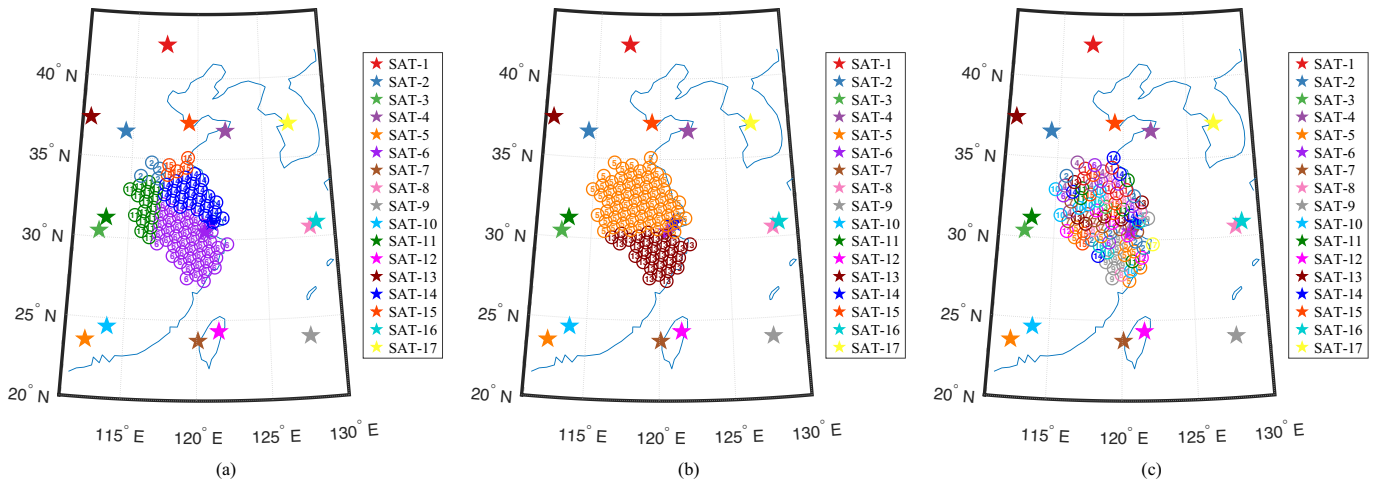


Fig. 6. Comparison of satellite-cell association results under (a) *MaxElevation*, (b) *MaxTime* and (c) *Proposed* schemes.

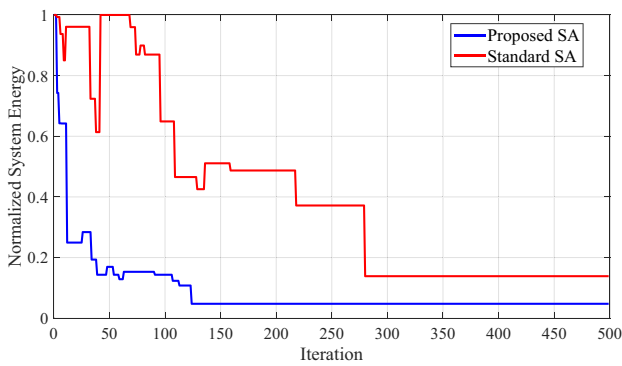


Fig. 7. Convergence of SA algorithm.

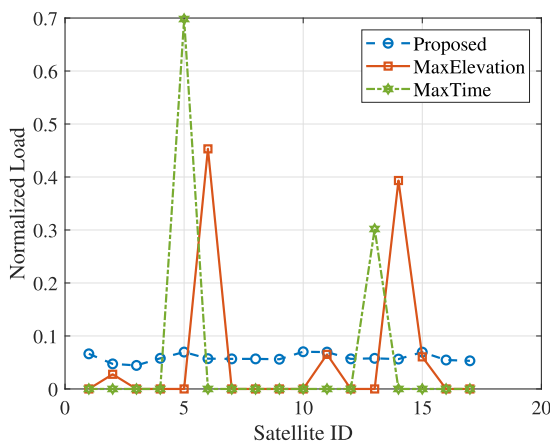


Fig. 8. Normalized traffic demand of all satellites under different satellite-cell association schemes.

mately 70% energy reduction within 30 iterations, a step-like optimization period for exploratory refinement, and a final stabilization phase after 120 iterations. In the standard SA, neighboring solutions are generated by randomly

selecting a cell and reassigning it to an arbitrary visible satellite. In terms of convergence speed, the standard SA requires about 280 iterations to reach a stable state, indicating a significantly slower convergence. Regarding optimization quality, the final normalized system energy of the standard SA remains around 0.14, while the improved SA further converges to 0.05, which is approximately 65% lower than that of the standard SA. These results collectively validate the superiority of our proposed multi-phase biased neighborhood generation mechanism.

To provide a clearer visualization of satellite load distribution, we define the normalized load as

$$\hat{L}_i = \frac{L_i}{\sum_{i=1}^{N_s} L_i}, \quad (18)$$

where $\hat{L}_i$ represents the normalized load of satellite i and L_i denotes the actual traffic load. Fig. 8 further compares the inter-satellite load disparities under different cell association strategies. Specifically, the *MaxTime* scheme incurs a substantial load gap of 0.70, while the *MaxElevation* strategy reduces it to 0.45. This is primarily because most traffic demands are concentrated on SAT 6 and 14, which are geographically closer to the target region as shown in Fig. 6a. In contrast, the proposed algorithm effectively reduces the inter-satellite load gap to 0.026, achieving improvements of approximately 96.3% and 94.2% over the *MaxTime* and *MaxElevation* schemes, respectively. Notably, the traffic loads on the previously overloaded SATs 6 and 14 are reduced by 86.7% and 84.6%, respectively.

4.2.2. Performance of system throughput and delay

To further evaluate the effectiveness of the proposed BH resource allocation algorithm, we conduct comprehensive simulations under varying traffic intensities based on the load-balanced satellite–cell association. The benchmark algorithms considered for performance comparison are outlined as follows.

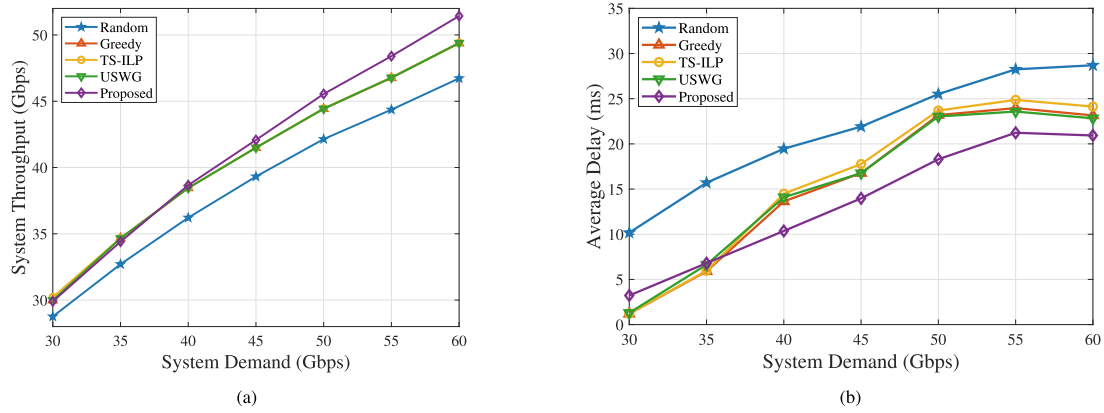


Fig. 9. Performance comparison of (a) system throughput and (b) average delay under different algorithms.

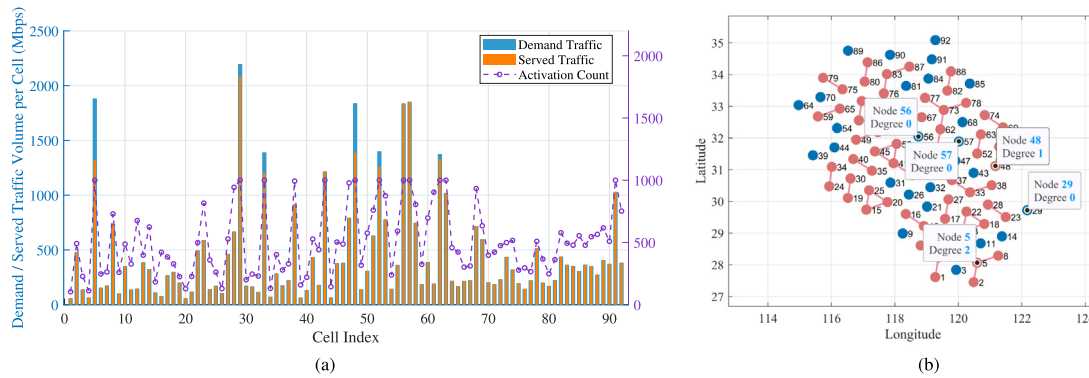


Fig. 10. The results of (a) comparison of traffic volume (demand vs. served) and beam activation count per cell during BH period with system traffic of 40 Gbps and (b) interference graph.

- *Random Beam Hopping (Random)*: In each time slot, three cells are randomly selected for service.
- *Greedy based Beam Hopping (Greedy)*: Cells with the highest traffic demands are prioritized and interfering cells are iteratively removed from the candidate set to satisfy interference constraints.
- *Two Step Integer Linear Programming Beam Hopping (TS-ILP)*: The illumination pattern is determined by solving two ILP problems sequentially. The first problem selects emergency service cells to maximize the aggregate demand, while the second addresses the remaining cells with the leftover beam resources. Both steps are subject to the beam-count limit and the interference constraint that no two selected cells are connected by an interference edge. A sliding-window-based satisfaction rate mechanism is incorporated to further enhance fairness and validate the effectiveness of the proposed method.
- *Maximum User Service Weight Gain Beam Hopping (USWG)*: Cell selection is guided by user service weights that reflect both traffic demand and delay urgency, following the methodology proposed by Wu et al. (2022).

In the above algorithms, all beams apply full-frequency reuse and CCI is mitigated through distance-based beam isolation. Power allocation for each beam is subsequently optimized using Lagrangian methods (Shu et al., 2022).

It's obvious that our proposed algorithm addresses the inherent limitation of full frequency reuse, which fails to simultaneously serve interfering cells. It introduces an interference-aware bandwidth allocation scheme and further employs a water-filling power allocation approach under non-uniform bandwidth conditions.

Fig. 9 illustrates the system throughput and average packet delay achieved by different algorithms under varying traffic demands ranging from 30 to 60 Gbps. As depicted in Fig. 9a, under low-traffic conditions, all benchmark algorithms (except *Random*) achieve comparable system throughput. As traffic intensity increases, the superiority of our proposed algorithm becomes increasingly evident. Despite not utilizing the full bandwidth on each beam, the proposed method achieves higher throughput by enabling more flexible and interference-aware resource allocation. At a total traffic demand of 60 Gbps, the proposed method achieves a throughput gain of 4%~10% over baseline schemes, confirming its superior spectrum efficiency.

In terms of delay performance shown in Fig. 9b, the proposed algorithm exhibits slightly higher latency than the *TS-ILP*, *USWG*, and *Greedy* schemes under light traffic conditions. This is primarily attributed to bandwidth fragmentation, which may reduce per-beam transmission capacity when resources are abundant. Nevertheless, as the traffic intensity increases, the proposed method effectively overcomes the limitations of traditional full-frequency reuse, which cannot simultaneously serve interfering cells. Through dynamic interference-aware bandwidth and power allocation, the proposed algorithm enables on-demand service and achieves a delay reduction of 16.15% to 32.64% compared to other algorithms under high-load scenarios. The slight decrease in delay observed at curve's extremity is attributed to packet drop caused by excessive load; since the reported metric is the average delay of successfully served packets, this phenomenon is reasonable and consistent with expectations.

4.2.3. Performance of traffic satisfaction rate

To provide a more comprehensive evaluation and further validate the effectiveness of the proposed algorithm, we present a detailed analysis of the statistical results across all cells under a representative moderately high traffic. Fig. 10a shows the demand and service traffic volume of each cell, as well as the number of times each cell is served by a beam during the entire BH scheduling period under our proposed algorithm. It can be observed that, except for a few hotspot cells, the traffic demands of most cells are well satisfied. Furthermore, the beam activation count per cell (purple dashed line) closely follows the trend of traffic demand, demonstrating that the algorithm is capable of allocating resources in a demand-aware manner.

Notably, among hotspot cells with comparable traffic volumes, noticeable differences in demand satisfaction rates can be observed. For instance, Cell 5 and Cell 48 achieve

relatively low satisfaction rates of only 70.0% and 75.4%, respectively, while Cell 29, Cell 56, and Cell 57 reach as high as 95.1% to 100%. The interference graph in Fig. 10b provides critical insight into this disparity. Cells 5 and 48 exhibit interference degrees of 2 and 1, respectively, sharing interference relationships with Cells 2, 8, and 58, all of which are also high-demand cells with traffic exceeding 500 Mbps. These cells require frequent service and bandwidth isolation due to mutual interference, leading to capacity degradation. In contrast, Cells 29, 56, and 57 are interference-free (i.e., degree 0) and can utilize the entire available bandwidth without constraints, thus achieving higher satisfaction rates.

To evaluate per-cell performance under high traffic demand (50 Gbps), Fig. 11 presents a comparative analysis of cell satisfaction rates across scheduling algorithms. As shown in Fig. 11a, the box plot illustrates service satisfaction rates across cells with the 10th to 90th percentile range and red crosses marking outliers. The proposed algorithm yields a significantly narrower box which indicates highly consistent satisfaction rates across cells. Notably, even the lowest outlier remains above 50% satisfaction, demonstrating significantly improved fairness. In contrast, although *TS-ILP* and *USWG* incorporate certain mechanisms to avoid starvation of low-demand cells, their fairness performance degrades under high traffic intensity, as reflected in the higher box and outliers with extremely low satisfaction rates. Fig. 11b shows the CDF of satisfaction rates. The proposed algorithm achieves a notable advantage, with 93.9% of cells reaching a satisfaction rate of at least 90%, surpassing the 86.8%–89.3% attained by baseline methods. This demonstrates that the proposed method not only improves average performance but also ensures that the majority of cells maintain high satisfaction levels, even under constrained bandwidth and interference conditions.

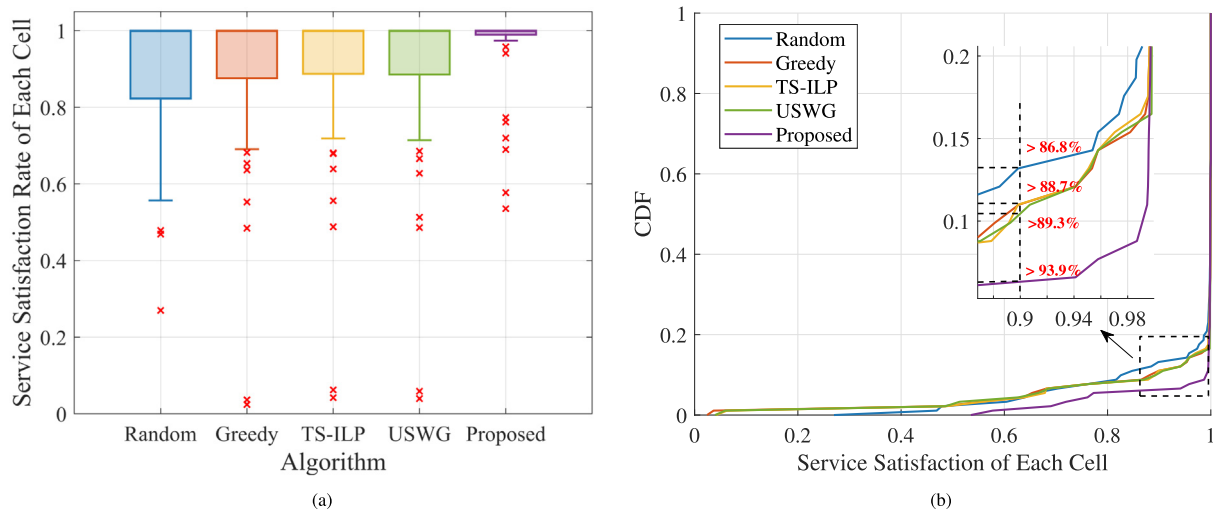


Fig. 11. Statistical analysis of cell satisfaction rate performance with system traffic of 50 Gbps: (a) Box plot (10th–90th percentiles) and (b) Cumulative Distribution Function (CDF) curve.

Table 2

Proportion of cells achieving a satisfaction rate exceeding 90 % for different algorithms under various traffic intensities.

Algorithm	Traffic intensity			
	30 Gbps	40 Gbps	50 Gbps	60 Gbps
Random	96.7 %	92.4 %	86.8 %	80.4 %
Greedy	100 %	93.5 %	88.7 %	84.8 %
TS-ILP	100 %	93.5 %	88.7 %	84.8 %
USWG	100 %	94.6 %	89.3 %	84.8 %
Proposed	100 %	95.7 %	93.9 %	89.1 %

To provide an overview of performance across varying traffic intensities, Table 2 summarizes the proportion of cells achieving satisfaction rates above 90% for each algorithm. As traffic intensity rises from 30 to 60 Gbps, all methods show a decline in this metric due to resource constraints. While most algorithms achieve near-perfect performance under light load, our method maintains superior service quality under medium-to-high traffic, demonstrating stronger tail control and superior inter-cell fairness even under network congestion.

4.2.4. Performance of average delay of each cell

Fig. 12 illustrates the statistical performance of the average packet response delay of each cell under different scheduling algorithms. The custom box plots in Fig. 12a (spanning 20th–80th percentiles) demonstrate that both the proposed algorithm and *USWG* achieve significantly lower and more compact delay distributions, indicating reduced delay with improved consistency across cells. Notably, *USWG* achieves superior delay performance compared to the proposed method, with a lower median delay of 1.42 ms versus 1.72 ms and a smaller upper-whisker value of 7.2 ms versus 9.1 ms. These improvements can be primarily attributed to its inherent delay-oriented scheduling mechanism. Although *USWG* performs better in terms of typical delay metrics, it results in a greater num-

ber of high-delay outliers. The proposed method reduces the number of cells with average delay exceeding 40 ms by approximately 43% compared to *USWG*, demonstrating enhanced fairness and service consistency.

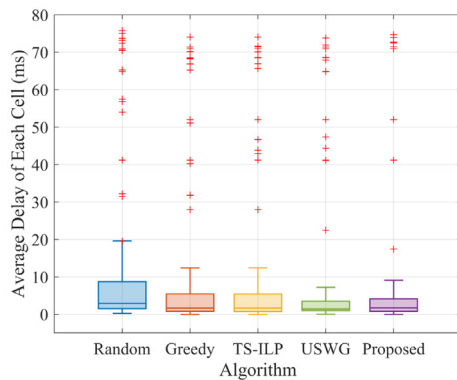
Fig. 12b presents the CDF curves of the average delay across all cells under different scheduling algorithms. It can be observed that the proposed method significantly outperforms the baselines in terms of low-latency coverage. Specifically, 90.2% of the cells under the proposed algorithm achieve an average delay below 10 ms, compared to 81.5–83.7% for the baseline methods, yielding an improvement of 7.8–10.7%. More importantly, the 90th percentile delay is reduced to 8.81 ms under the proposed method, compared to 52.50 ~ 65.35 ms under baseline algorithms, representing a reduction of 83.2–86.5% in tail latency.

Table 3 summarizes the 90th percentile of average packet delays in all cells for each algorithm across varying traffic intensities. At a low traffic demand of 30 Gbps, *USWG* achieves the lowest delay by employing an explicit delay-aware scheduling mechanism, confining delays below 2 ms for most cells. As intensity increases, the proposed algorithm's superiority emerges prominently in controlling tail latencies for high-delay cells. While baseline methods suffer from a sharp rise in the 90th percentile delay due to numerous high-delay outliers, our approach maintains a compact delay distribution, effectively mitigating this

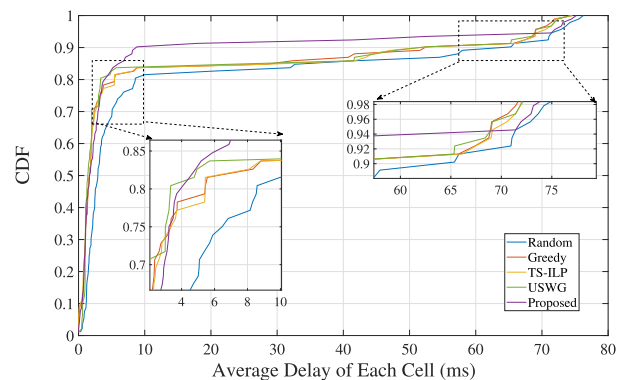
Table 3

Comparison of the 90th percentile of average delay for all cells under different traffic intensities.

Algorithm	Traffic intensity			
	30 Gbps	40 Gbps	50 Gbps	60 Gbps
Random	7.9 ms	26.4 ms	65.3 ms	71.1 ms
Greedy	3.1 ms	31.7 ms	52.5 ms	70.6 ms
TS-ILP	3.1 ms	40.5 ms	52.5 ms	70.6 ms
USWG	2.0 ms	41.7 ms	52.5 ms	70.6 ms
Proposed	4.1 ms	6.8 ms	8.8 ms	68.3 ms



(a)



(b)

Fig. 12. Comparison of average delay performance under different algorithms with system traffic of 50 Gbps: (a) box plot (20th–80th percentiles); (b) CDF curve.

degradation and confirming its superiority in ensuring inter-cell service consistency.

5. Conclusion

This paper tackles the problem of resource scheduling in multi-satellite beam hopping systems to efficiently match multi-dimensional resources with spatiotemporally uneven traffic demands. The originally formulated NP-hard problem is sequentially decomposed into three tractable subproblems. First, leveraging the multi-coverage property of NGSO constellations, a satellite–cell association strategy based on simulated annealing was developed to balance the traffic load across satellites. Subsequently, the illumination pattern was determined by maximizing the aggregate traffic demand, upon which a graph-based interference-aware algorithm flexibly assigns frequency bands to avoid co-channel interference. Finally, beam powers are further refined based on available bandwidth and traffic demand to realize efficient on-demand service. Extensive simulations verify the superiority of the proposed scheme in enhancing system throughput, reducing average delay and load gap, while achieving consistent service fairness across ground cells.

While the proposed framework demonstrates clear advantages as an integrated solution, we acknowledge that a comprehensive and standalone analysis of each constituent algorithm lies beyond the scope of this system-oriented article. Several promising directions are therefore outlined for future work. We intend to conduct thorough and focused studies on each subproblem, advancing the theoretical modeling and rigorously benchmarking the proposed methods against existing alternatives in terms of key performance indicators such as computational time, solution quality, complexity, and stability. Furthermore, inter-satellite handover mechanisms and their seamless integration into the scheduling framework will be explored to enhance operational continuity and mitigate service interruptions in practical large-scale deployments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of the China Youth Project under Grant No. 12104485.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.asr.2025.12.079>.

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